EVPN/Overlay and Network Architecture

Satoru Matsushima



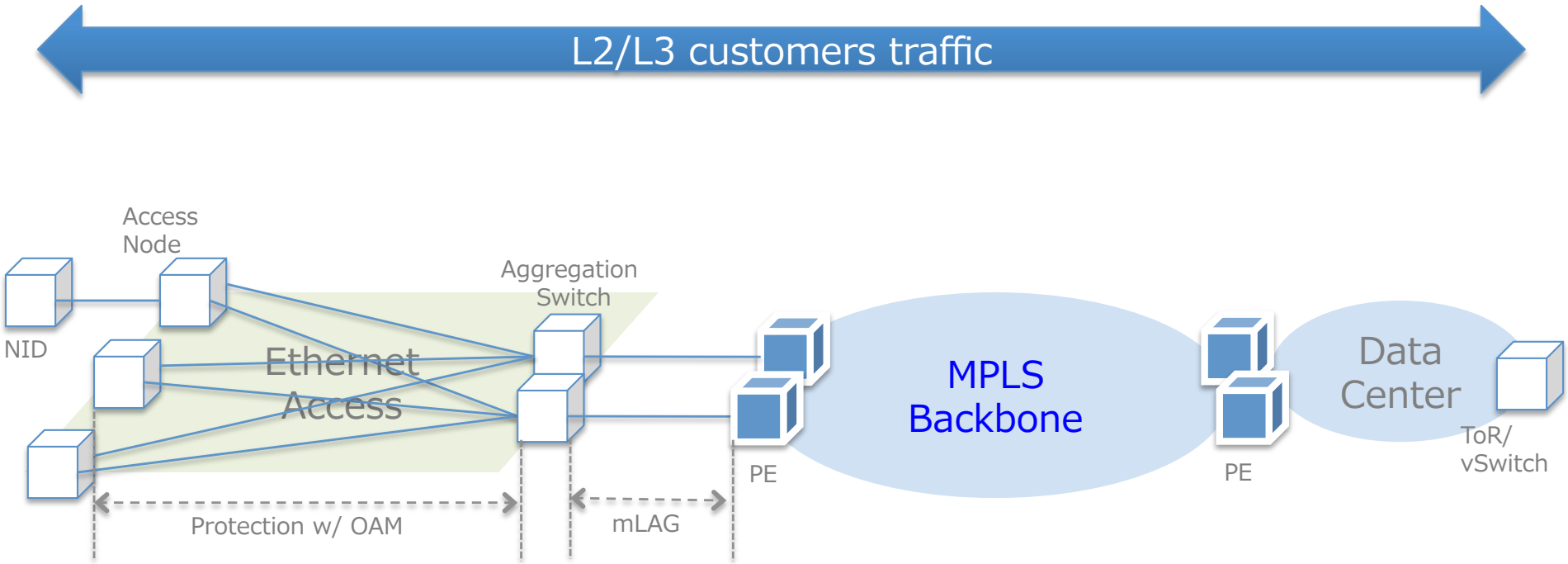
What Networks We Have…

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An E2E Architectural View of Network Service

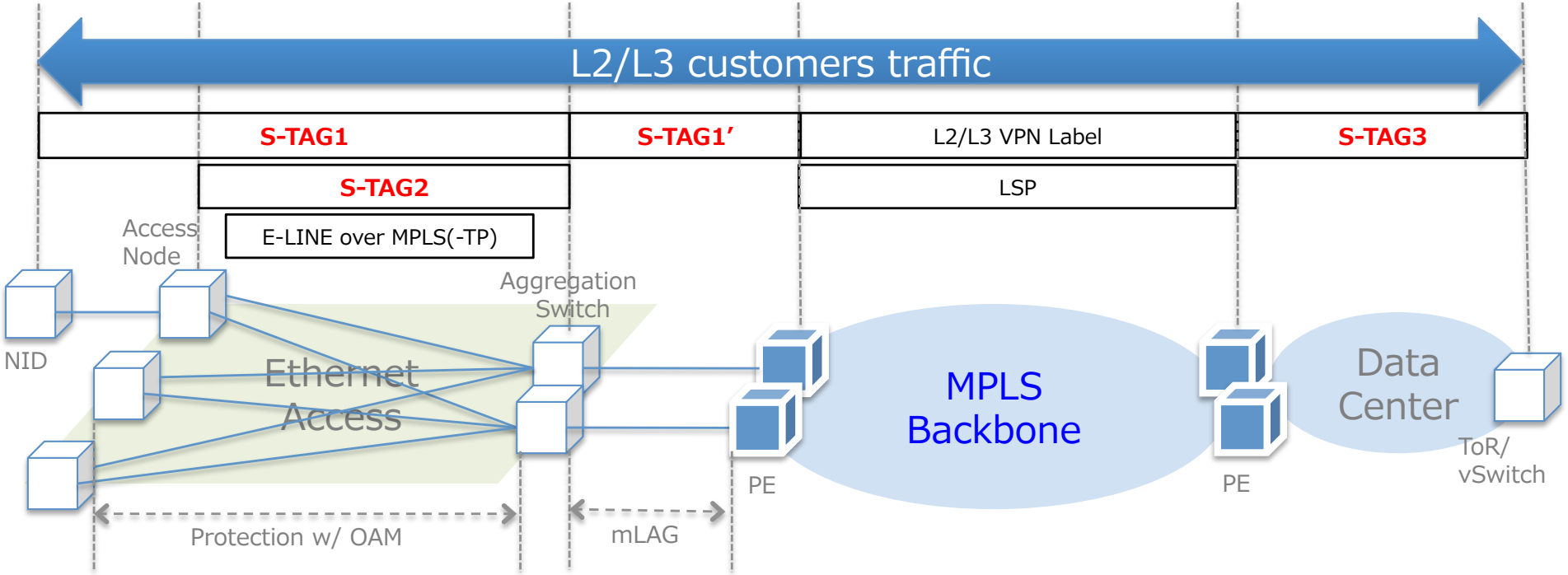
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An E2E Architectural View of Network Service

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VLAN Stack Aggregation on Access and Data-Center networks



Issues of Tag-stacking based Ethernet Access

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■VID Scalability

- 0-4096 (12bits space)
- Stacking S-TAG works to enlarge VID space (12+12=24bits)
- But aggregation switches still need to be deployed for each 12bits VID space

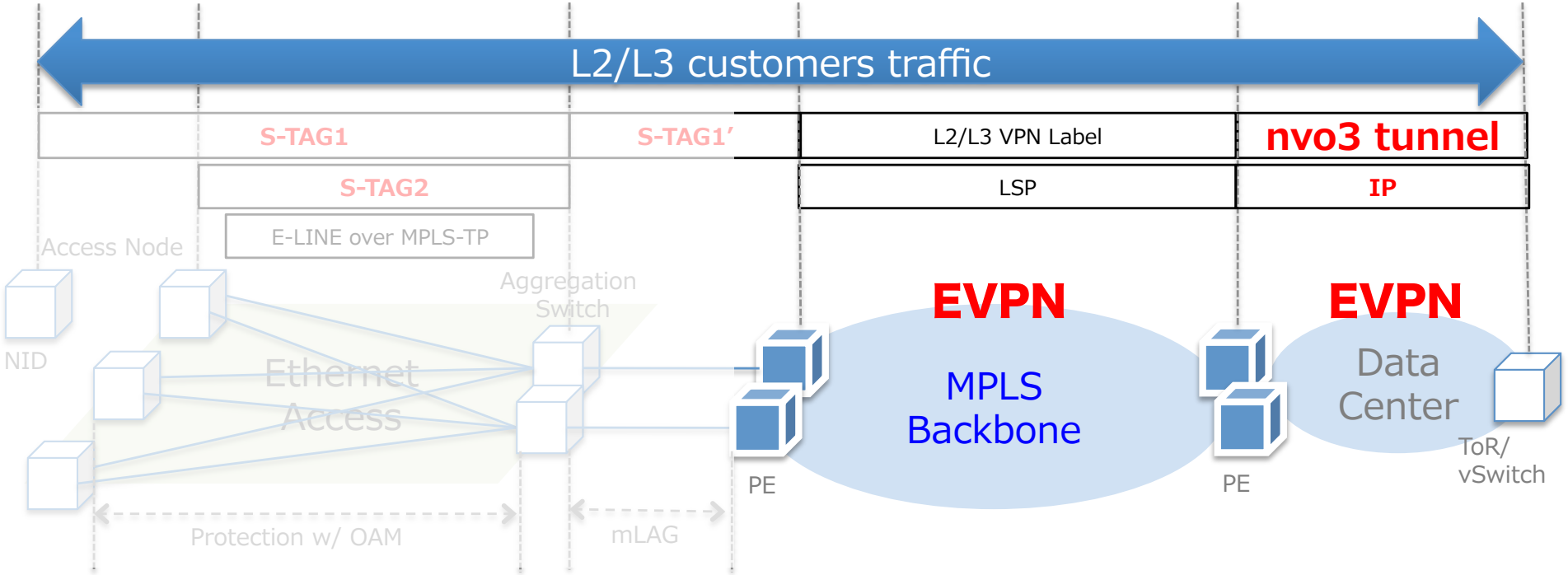
■Deployment pains

- Agg-switches become single point of failure that are increased as network growth
- Deploying 1:1 Agg-switch redundancy works to spend more money

Getting over the issues of Ethernet networking

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Emerging NVO3 and EVPN Control-plane to solve the issues



VPLS vs EVPN

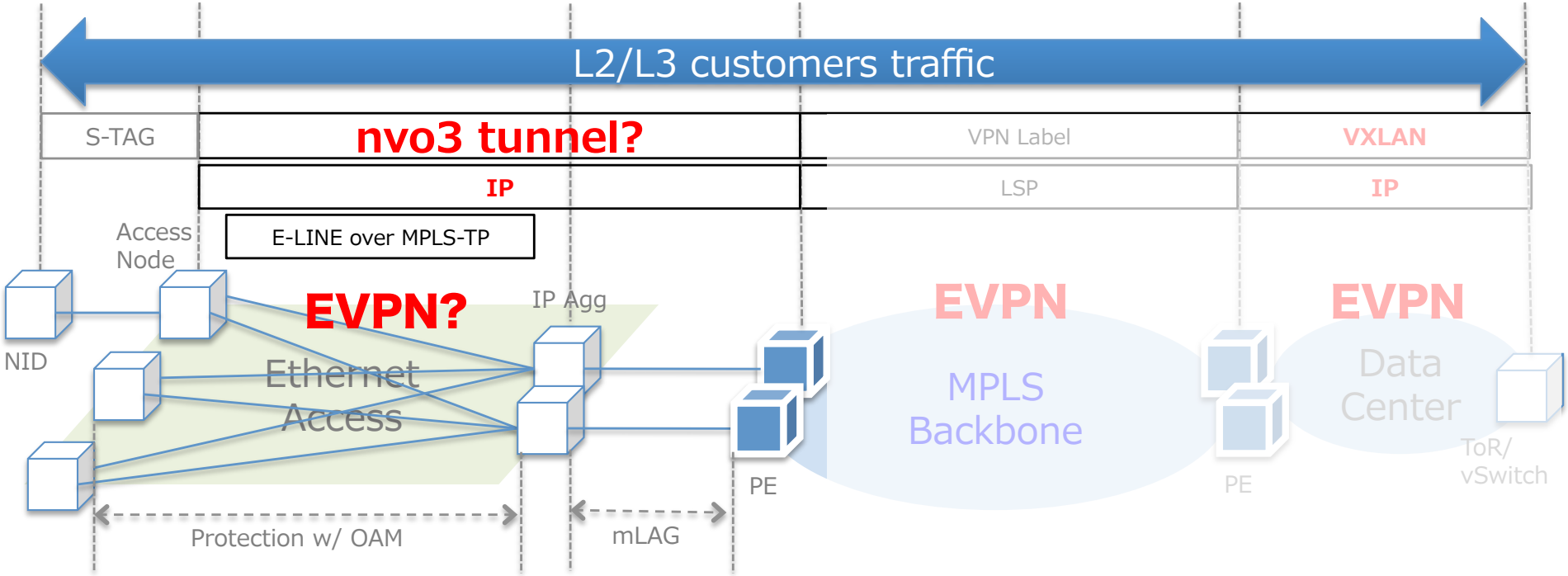
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	VPLS（RFC4762）	EVPN（RFC7432）
VPN Signaling	LDP	BGP
MAC Learning	Data-Plane	BGP Policy
PE-CE Multi-homing	Single-Active	All-Active
Convergence	MAC Re-Learn	BGP Update
BUM	Ingress copy	Multicast

Getting over the issues of Ethernet networking

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Can NVO3 relief the pains of Ethernet Access as well?



Challenges to deploy NVO3 into the Access

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- Access-node might not be EVPN capable
  - Due to limited resources to reduce cost
  - No MAC learning
  - Is that Softbank specific?

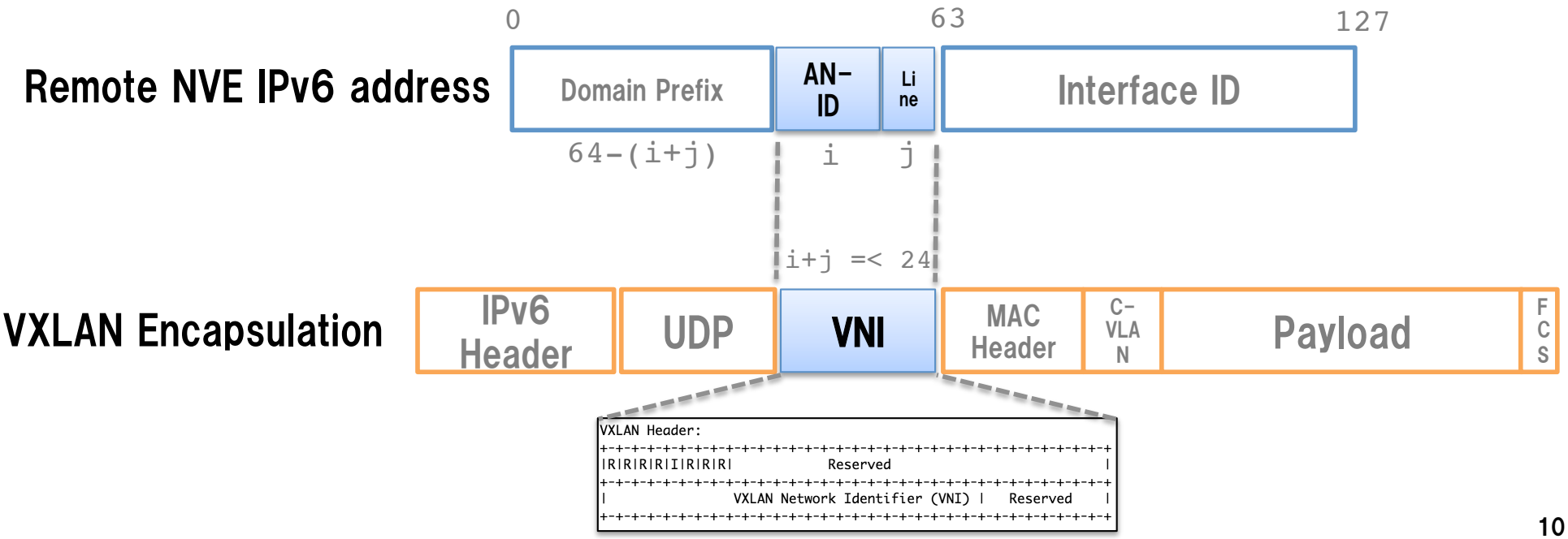
- How does the PE find remote NVEs without BGP?
  - PPP/L2TP Signaling per tunnel? -> NO!
  - Using controller? Is that scalable for 10K~ ANs deployment?

- Other issues
  - Which encapsulation should be chose, e.g., VXLAN, GRE, etc.,
  - Multicast?

IPv6 might help for NVE discovery

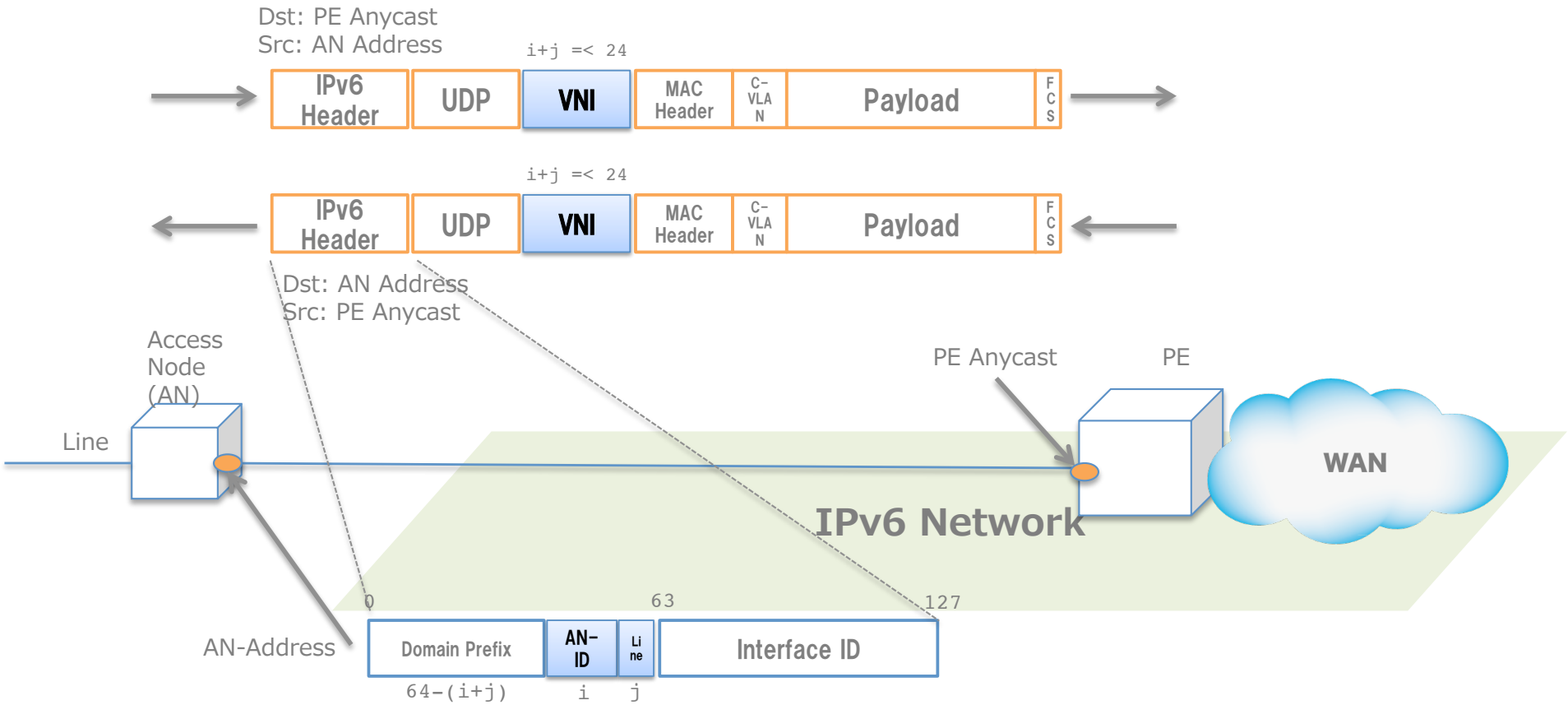
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Remote NVE address can be derived from VNID with mapped IPv6 prefix



Packet Flow and Format

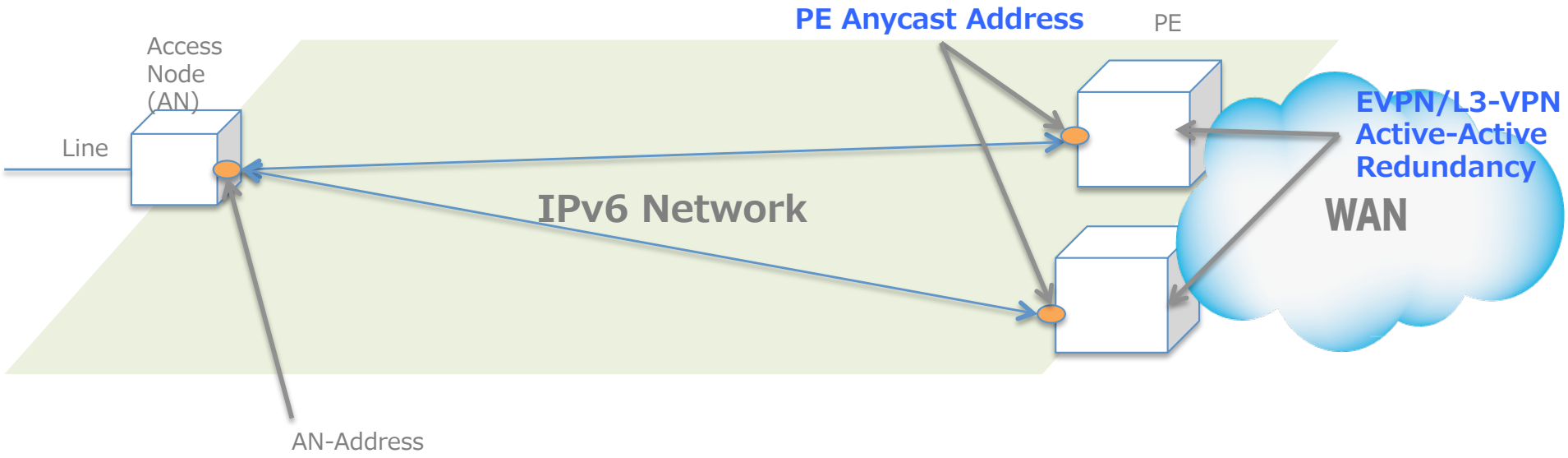
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Multi-homing w/ Anycast routing*

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More Resilient Connectivity than VLAN/Pseudo-Wire

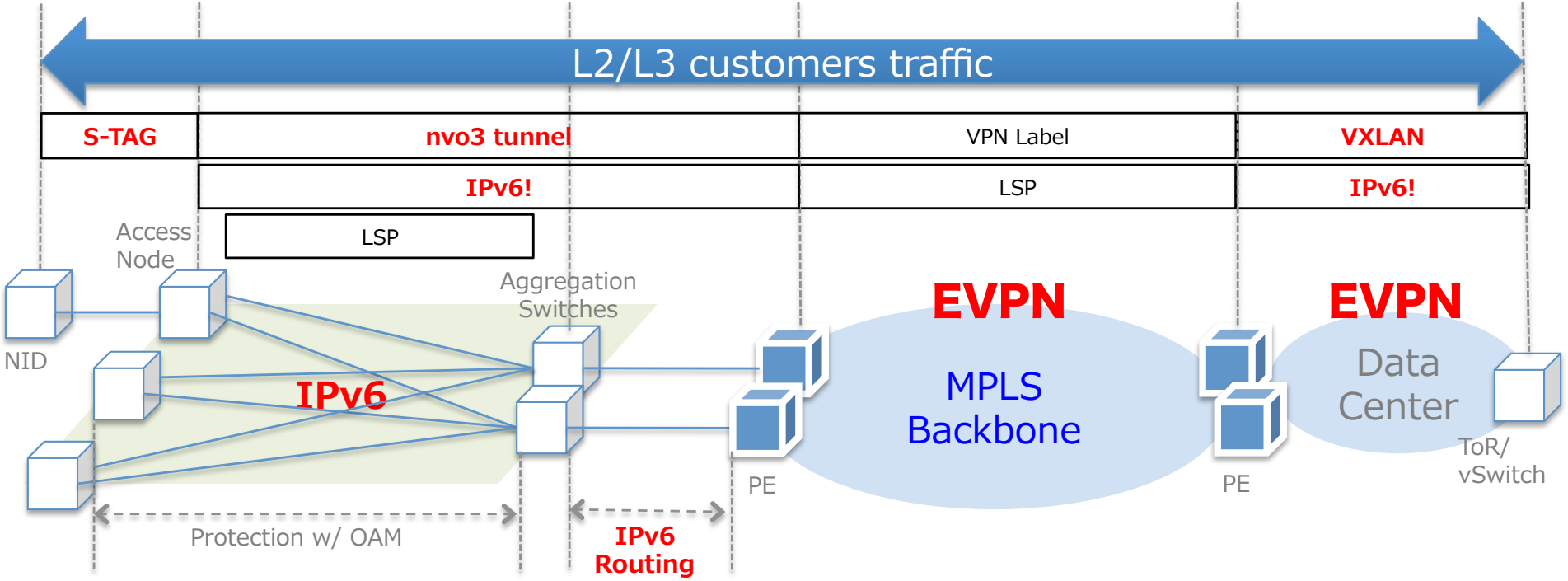


* https://tools.ietf.org/html/draft-sajassi-bess-pbb-evpn-anycast-ip-tunnels-00

The Ambitious Network in the future

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NVO3 Ethernet Access over IPv6-only Access Network



Summary

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- Pains exist not only in DC, but also in Access
- NVO3 can help to relief the pain, but…
  - It depends on node's capability and resource restriction
- Utilizing IPv6 shall suite "Light Weight" overlay deployment for Ethernet Access
  - without not only BGP but also any other signaling
- It's Controversial for changing VNID semantics?
  - Don't worry, MEF already did it for VID in E-Line. ☺

Questions? Comments?

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